EXPERIMENT 2 – DATA DISCOVERY AND PREPARATION

AIM

To explore the Telco churn data, detect outliers using IQR, investigate churn patterns, split the dataset 80:20 and save a processed CSV.

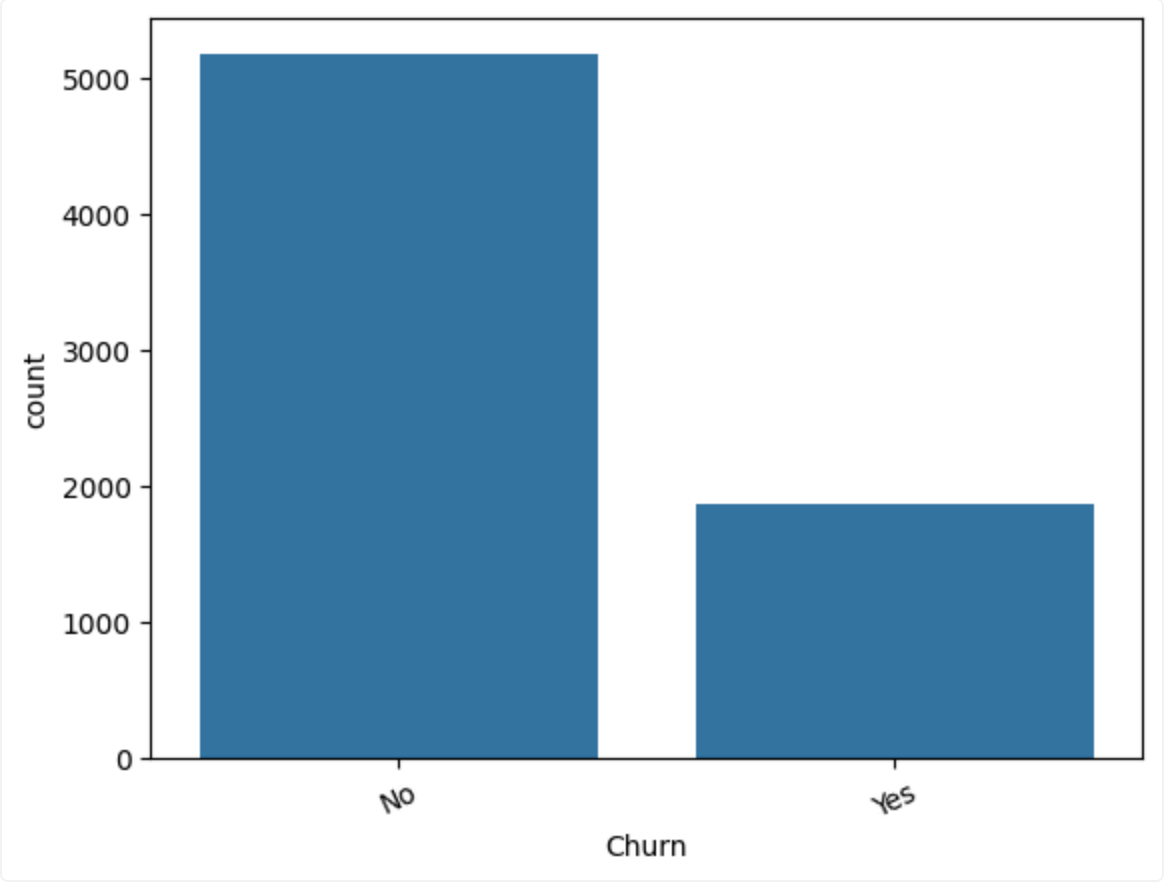
1. Load and clean the Telco dataset.
2. Plot distributions for Churn, gender, Contract and PaymentMethod.
3. Use histograms and box plots for numeric columns.
4. Detect outliers using IQR bounds Q1−1.5×IQR and Q3+1.5×IQR.
5. Compare churn with Contract, InternetService, MonthlyCharges and tenure.
6. Encode input features, split into train/test data and save the processed CSV.

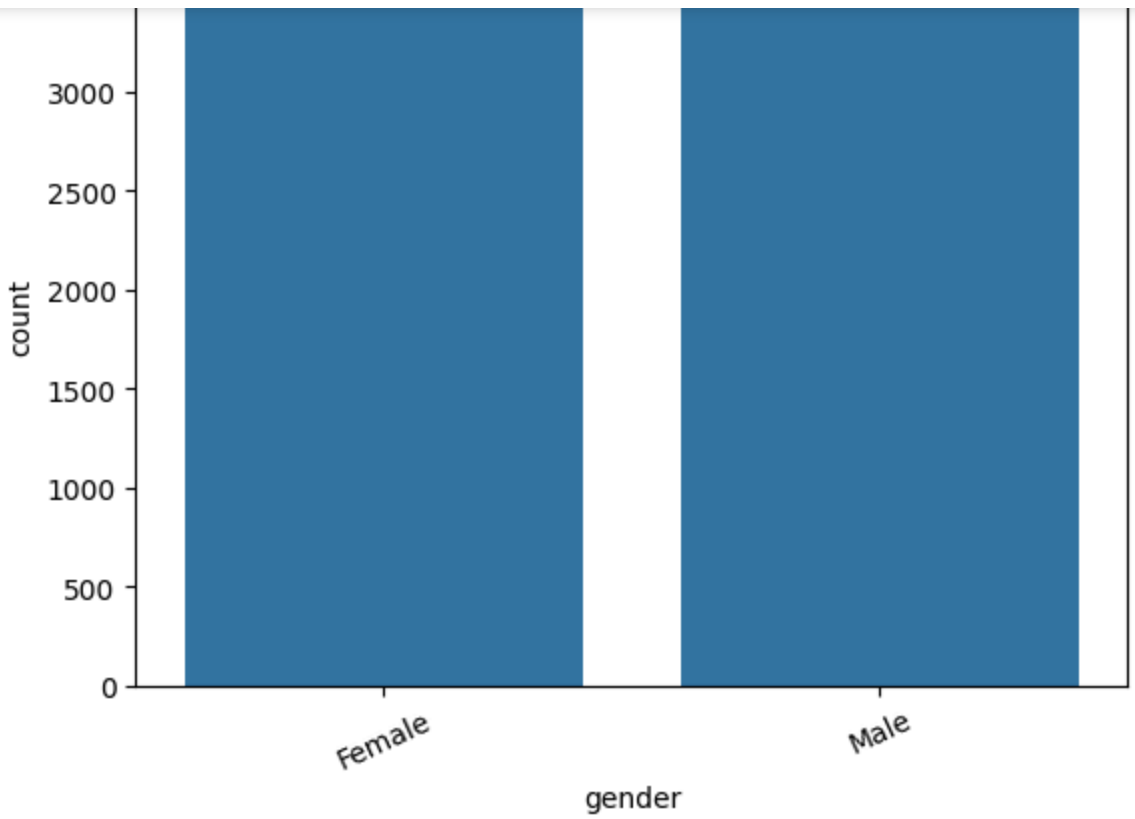
In [ ]

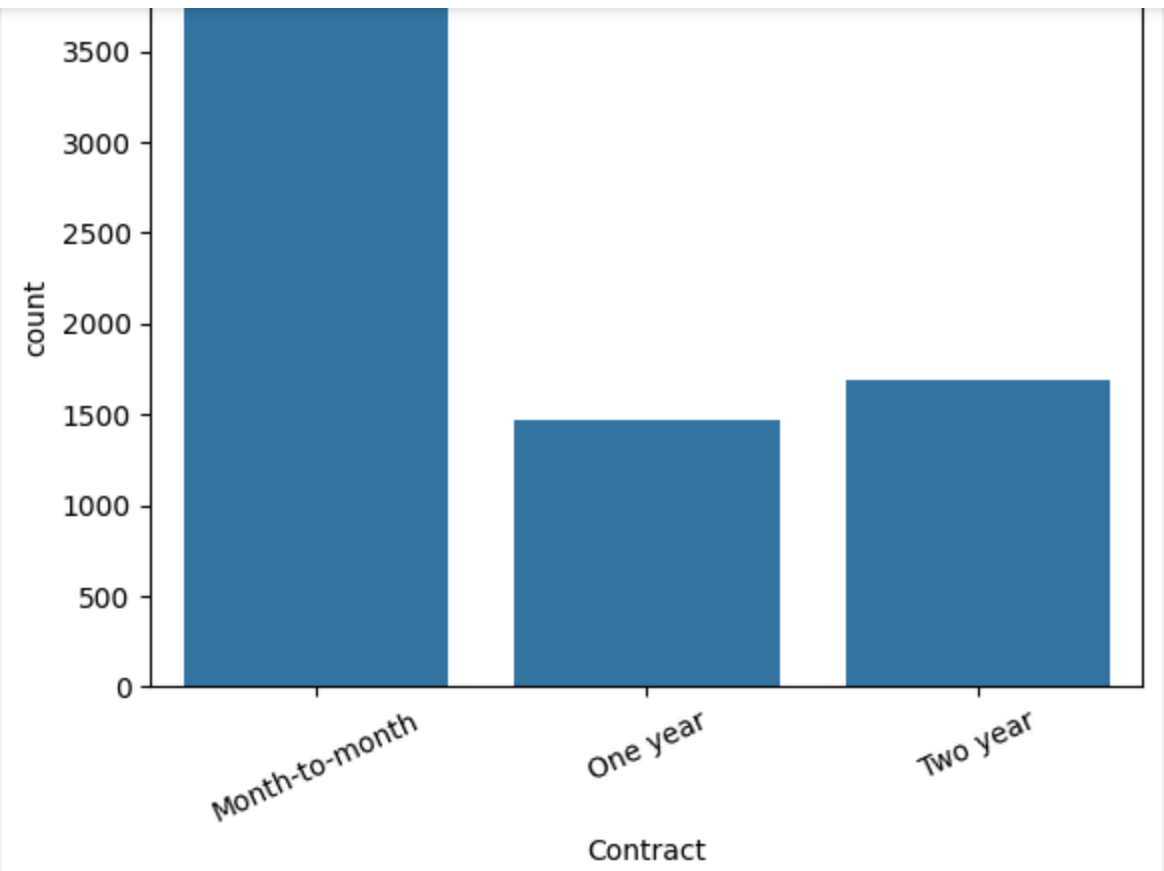
Copy

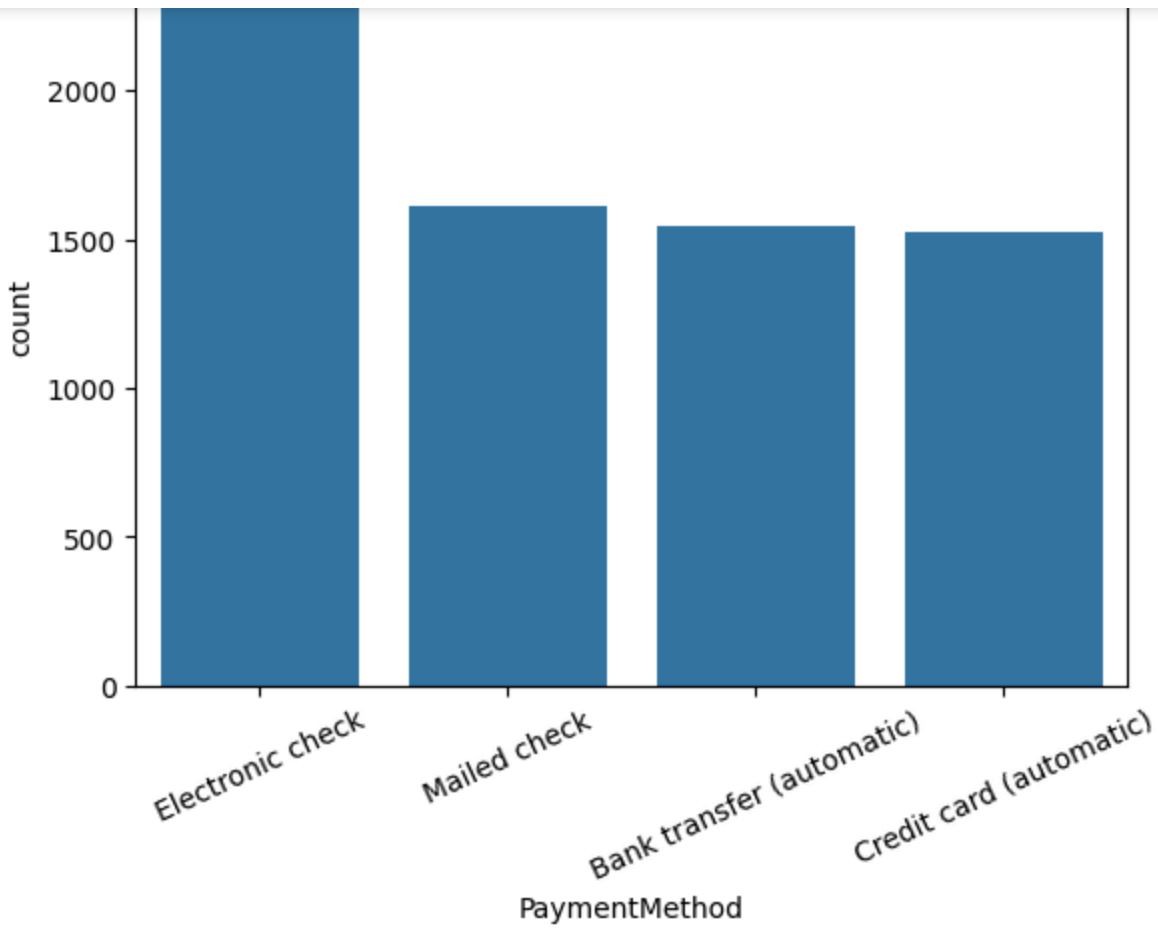
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split

df = pd.read_csv("Telco-Customer-Churn.csv.csv").drop_duplicates().co
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], erro
for c in df.select_dtypes(include="number"):
    df[c] = df[c].fillna(df[c].median())
for c in df.select_dtypes(exclude="number"):
    if df[c].isnull().any(): df[c] = df[c].fillna(df[c].mode()[0])
for c in ["Churn","gender","Contract","
    sns.countplot(data=df,x=c); plt.xticks(rotation=25); plt.show()
for c in ["tenure","MonthlyCharges","TotalCharges
    sns.histplot(df[c],kde=True); plt.show(); sns.boxplot(x=df[c]); plt.show()
    q1,q3=df[c].quantile([.25,.75]); iqr=q3-q1
    print(c,"IQR outliers:",((df[c]<q1-1.5*iqr)|(df[c]>q3+1.5*iqr)).s
for c in ["Contract","InternetService"]:
    sns.countplot(data=df,x=c,hue="Churn"); plt.xticks(rotation=25);
sns.boxplot(data=df,x="Churn",y="MonthlyCharges"); plt.show
sns.boxplot(data=df,x="Churn",y="tenure"); plt.show()
y=df["Churn"].map({"No":0,"Yes":1})
X=pd.get_dummies(df.drop(columns=["Churn","customerID"],err
Xtr,Xte,ytr,yte=train_test_split(X,y,test_size=.2,random_state=42,stratify=y)
processed=X.copy(); processed["Churn"]=y
```


5000
4000
3000
2000
1000
0
count
No
Yes
Churn


3000
2500
2000
1500
1000
500
0
count
Female
Male
gender


3500
3000
2500
2000
1500
1000
500
0
count
Month-to-month
One year
Two year
Contract



KeyboardInterrupt

```
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipykernel_685/3132256051.py in <cell line: 0>()
     13     sns.countplot(data=df,x=c); plt.xticks(rotation=25); plt.show()
     14 for c in ["tenure","MonthlyCharges","TotalCharges"]:
---> 15     sns.histplot(df[c],kde=True); plt.show(); sns.boxplot(x=df[c]);
plt.show()
     16     q1,q3=df[c].quantile([.25,.75]); iqr=q3-q1
     17     print(c,"IQR outliers:",((df[c]<q1-1.5*iqr)|
(df[c]>q3+1.5*iqr)).sum())

/usr/local/lib/python3.13/dist-packages/matplotlib/pyplot.py in show(*args,
**kwargs)
    612     """
    613     _warn_if_gui_out_of_main_thread()
--> 614     return _get_backend_mod().show(*args, **kwargs)
    615
```

```
show(close, block)
     88     try:
     89         for figure_manager in Gcf.get_all_fig_managers():
---> 90             display(
     91                 figure_manager.canvas.figure,
     92
metadata=_fetch_figure_metadata(figure_manager.canvas.figure),

/usr/local/lib/python3.13/dist-packages/IPython/core/display.py in
display(include, exclude, metadata, transient, display_id, *objs, **kwargs)
    318             publish_display_data(data=obj, metadata=metadata, **kwargs)
    319         else:
--> 320             format_dict, md_dict = format(obj, include=include,
exclude=exclude)
    321             if not format_dict:
    322                 # nothing to display (e.g. _ipython_display_ took over)

/usr/local/lib/python3.13/dist-packages/IPython/core/formatters.py in
format(self, obj, include, exclude)
    178             md = None
    179             try:
--> 180                 data = formatter(obj)
    181             except:
    182                 # FIXME: log the exception

<decorator-gen-2> in __call__(self, obj)

/usr/local/lib/python3.13/dist-packages/IPython/core/formatters.py in
catch_format_error(method, self, *args, **kwargs)
    222     """show traceback on failed format call"""
    223     try:
--> 224         r = method(self, *args, **kwargs)
    225     except NotImplementedError:
    226         # don't warn on NotImplementedErrors

/usr/local/lib/python3.13/dist-packages/IPython/core/formatters.py in
__call__(self, obj)
    339                 pass
    340             else:
--> 341                 return printer(obj)
    342             # Finally look for special method names
    343             method = get_real_method(obj, self.print_method)
```

```
    150
--> 151     fig.canvas.print_figure(bytes_io, **kw)
    152     data = bytes_io.getvalue()
    153     if fmt == 'svg':

/usr/local/lib/python3.13/dist-packages/matplotlib/backend_bases.py in
print_figure(self, filename, dpi, facecolor, edgecolor, orientation, format,
bbox_inches, pad_inches, bbox_extra_artists, backend, **kwargs)
   2182                 # force the figure dpi to 72), so we need to set it
again here.
   2183                 with cbook._setattr_cm(self.figure, dpi=dpi):
-> 2184                     result = print_method(
   2185                         filename,
   2186                         facecolor=facecolor,

/usr/local/lib/python3.13/dist-packages/matplotlib/backend_bases.py in <lambda>
(*args, **kwargs)
   2038                 "bbox_inches_restore"}
   2039             skip = optional_kws - {*inspect.signature(meth).parameters}
-> 2040             print_method = functools.wraps(meth)(lambda *args,
**kwargs: meth(
   2041                 *args, **{k: v for k, v in kwargs.items() if k not in
skip}))
   2042         else:  # Let third-parties do as they see fit.

/usr/local/lib/python3.13/dist-packages/matplotlib/backends/backend_agg.py in
print_png(self, filename_or_obj, metadata, pil_kwargs)
    479             *metadata*, including the default 'Software' key.
    480         """
--> 481         self._print_pil(filename_or_obj, "png", pil_kwargs, metadata)
    482
    483     def print_to_buffer(self):

/usr/local/lib/python3.13/dist-packages/matplotlib/backends/backend_agg.py in
_print_pil(self, filename_or_obj, fmt, pil_kwargs, metadata)
    427         *pil_kwargs* and *metadata* are forwarded).
    428         """
--> 429         FigureCanvasAgg.draw(self)
    430         mpl.image.imsave(
    431             filename_or_obj, self.buffer_rgba(), format=fmt,
origin="upper",

/usr/local/lib/python3.13/dist-packages/matplotlib/backends/backend_agg.py in
```

```
⟶ 382             self.figure.draw(self.renderer)
    383             # A GUI class may be need to update a window using this
draw, so
    384             # don't forget to call the superclass.

/usr/local/lib/python3.13/dist-packages/matplotlib/artist.py in
draw_wrapper(artist, renderer, *args, **kwargs)
     92     @wraps(draw)
     93     def draw_wrapper(artist, renderer, *args, **kwargs):
---> 94         result = draw(artist, renderer, *args, **kwargs)
     95         if renderer._rasterizing:
     96             renderer.stop_rasterizing()

/usr/local/lib/python3.13/dist-packages/matplotlib/artist.py in
draw_wrapper(artist, renderer)
     69                 renderer.start_filter()
     70
---> 71             return draw(artist, renderer)
     72         finally:
     73             if artist.get_agg_filter() is not None:

/usr/local/lib/python3.13/dist-packages/matplotlib/figure.py in draw(self,
renderer)
   3255
   3256                 self.patch.draw(renderer)
→ 3257                 mimage._draw_list_compositing_images(
   3258                     renderer, self, artists, self.suppressComposite)
   3259

/usr/local/lib/python3.13/dist-packages/matplotlib/image.py in
_draw_list_compositing_images(renderer, parent, artists, suppress_composite)
    132     if not_composite or not has_images:
    133         for a in artists:
⟶ 134             a.draw(renderer)
    135     else:
    136         # Composite any adjacent images together

/usr/local/lib/python3.13/dist-packages/matplotlib/artist.py in
draw_wrapper(artist, renderer)
     69                 renderer.start_filter()
     70
---> 71             return draw(artist, renderer)
     72         finally:
```

```
renderer)
   3179             _draw_rasterized(self.get_figure(root=True),
artists_rasterized, renderer)
   3180
→ 3181         mimage._draw_list_compositing_images(
   3182             renderer, self, artists,
self.get_figure(root=True).suppressComposite)
   3183

/usr/local/lib/python3.13/dist-packages/matplotlib/image.py in
_draw_list_compositing_images(renderer, parent, artists, suppress_composite)
    132     if not_composite or not has_images:
    133         for a in artists:
——> 134             a.draw(renderer)
    135     else:
    136         # Composite any adjacent images together

/usr/local/lib/python3.13/dist-packages/matplotlib/artist.py in
draw_wrapper(artist, renderer)
     69                 renderer.start_filter()
     70
---> 71             return draw(artist, renderer)
     72         finally:
     73             if artist.get_agg_filter() is not None:

/usr/local/lib/python3.13/dist-packages/matplotlib/axis.py in draw(self,
renderer)
   1423         self.label.draw(renderer)
   1424
→ 1425         self._update_offset_text_position(tlb1, tlb2)
   1426         self.offsetText.set_text(self.major.formatter.get_offset())
   1427         self.offsetText.draw(renderer)

/usr/local/lib/python3.13/dist-packages/matplotlib/axis.py in
_update_offset_text_position(self, bboxes, bboxes2)
   2485                 bottom = self.axes.bbox.ymin
   2486             else:
→ 2487                 bbox = mtransforms.Bbox.union(bboxes)
   2488                 bottom = bbox.y0
   2489             y = bottom - self.OFFSETTEXTPAD *
self.get_figure(root=True).dpi / 72

/usr/local/lib/python3.13/dist-packages/matplotlib/transforms.py in
```

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⟶ 648          y0 = np.min([bbox.ymin for bbox in bboxes])
    649          y1 = np.max([bbox.ymax for bbox in bboxes])
    650          return Bbox([[x0, y0], [x1, y1]])

/usr/local/lib/python3.13/dist-packages/matplotlib/transforms.py in ymin(self)
    305     def ymin(self):
    306         """The bottom edge of the bounding box."""
⟶ 307         return np.min(self.get_points()[:, 1])
    308
    309     @property

/usr/local/lib/python3.13/dist-packages/numpy/_core/fromnumeric.py in min(a,
axis, out, keepdims, initial, where)
   3341     6
   3342     """
→ 3343     return _wrapreduction(a, np.minimum, 'min', axis, None, out,
   3344                           keepdims=keepdims, initial=initial,
where=where)
   3345

/usr/local/lib/python3.13/dist-packages/numpy/_core/fromnumeric.py in
_wrapreduction(obj, ufunc, method, axis, dtype, out, **kwargs)
     84                 return reduction(axis=axis, out=out, **passkwargs)
     85
---> 86     return ufunc.reduce(obj, axis, dtype, out, **passkwargs)
     87
     88

KeyboardInterrupt:
```

RESULT

Thus, churn patterns and outliers are explored, and the processed dataset and 80:20 split are produced.